

Welcome to NDNcomm 2015

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UCLA School of Theater, Film and Television

The Future of the Internet
is the Future of Storytelling

Storytelling is the Last Mile in **Big Data and Analytics**



Stanford SOCIAL INNOVATION REVIEW

Informing and inspiring leaders of social change

Organizational Development

Building a Storytelling Culture

Forbes / Entrepreneurs

SEP 17, 2015 @ 10:30 AM 1,017 VIEWS

Here's How Interactive Content Is Growth-Hacking Brands' Storytelling Abilities



THE 4 I'S OF STORYTELLING: OUT OF THE SCREEN, INTO MY WORLD

IMMERSION

Can I go deeper into the story-world, by learning more about it or by heightening my sensory experience of it?



INTERACTIVITY

Can I change or influence elements of the story?
Can I interact with other people around the story?



INTEGRATION

Is a cohesive story being told across platforms?
Can it interface with the real world in any way?



IMPACT

Does the story inspire me to take action in my own life, such as making a purchase or supporting a good cause?



"C2MTL [is] a hugely inspiring and entertaining conference that explores the intersection between commerce and creativity – something that has always been central to the Virgin brand."

(Sir Richard Branson - Founder, Virgin Group, Speaker at C2MTL, 2015)



C2MTL 2015 - London - 10/10/15

THE WALL STREET JOURNAL.

CIO Journal

Exclusive reporting and analysis for corporate-technology executives.

Technology, Marketing and Storytelling



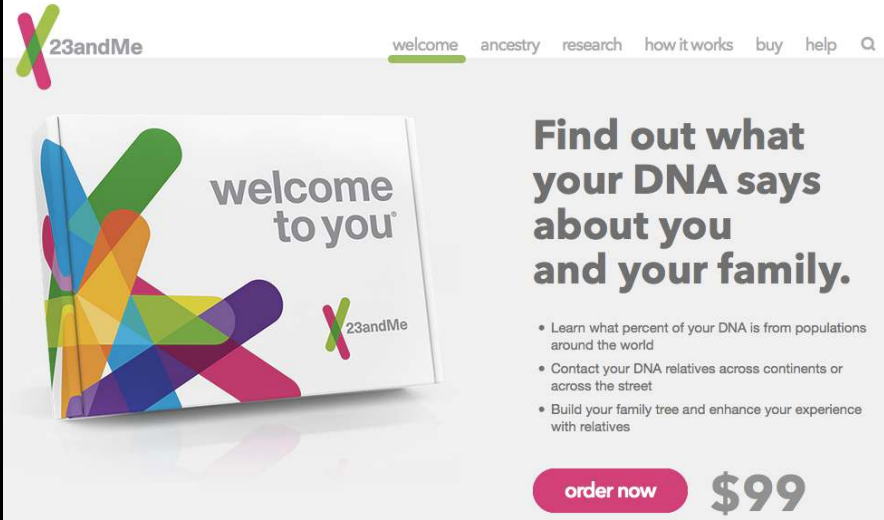
SCHEDULE



Stories Asunder: Tales for the Internet of Things

Personal Health and Wellness

A new medical history
based on personal
genetics and exposure,
as well as epigenetics.



The 23andMe website features a clean, modern design. The header includes the 23andMe logo and navigation links for welcome, ancestry, research, how it works, buy, help, and a search icon. The main banner displays a colorful, abstract graphic of overlapping shapes in green, blue, orange, and purple, with the text "welcome to you" and the 23andMe logo. To the right, a headline reads "Find out what your DNA says about you and your family." Below this, three bullet points list features: "Learn what percent of your DNA is from populations around the world," "Contact your DNA relatives across continents or across the street," and "Build your family tree and enhance your experience with relatives." A prominent pink button labeled "order now" is positioned next to the price "\$99".

23andMe

welcome ancestry research how it works buy help

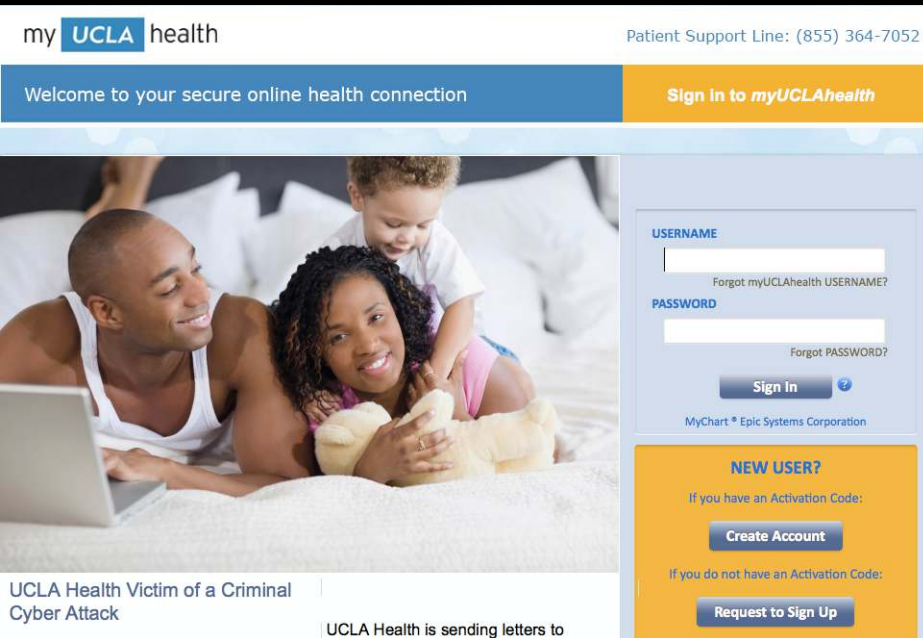
welcome to you

23andMe

Find out what your DNA says about you and your family.

- Learn what percent of your DNA is from populations around the world
- Contact your DNA relatives across continents or across the street
- Build your family tree and enhance your experience with relatives

order now **\$99**



The myUCLAhealth website has a blue and orange color scheme. The header includes the myUCLAhealth logo and the Patient Support Line: (855) 364-7052. Below the header, a blue bar contains the text "Welcome to your secure online health connection" and an orange bar contains the text "Sign in to myUCLAhealth". The main content area features a large image of a family (a man, a woman, and a child) lying in bed, with the man using a laptop. To the right of the image is a login form with fields for USERNAME and PASSWORD, each with a "Forgot myUCLAhealth USERNAME?" or "Forgot PASSWORD?" link. Below the fields is a "Sign In" button and the text "MyChart® Epic Systems Corporation". Below the login form is a "NEW USER?" section with two options: "If you have an Activation Code:" with a "Create Account" button, and "If you do not have an Activation Code:" with a "Request to Sign Up" button.

my **UCLA** health

Patient Support Line: (855) 364-7052

Welcome to your secure online health connection

Sign in to myUCLAhealth

USERNAME

Forgot myUCLAhealth USERNAME?

PASSWORD

Forgot PASSWORD?

Sign In

MyChart® Epic Systems Corporation

NEW USER?

If you have an Activation Code:

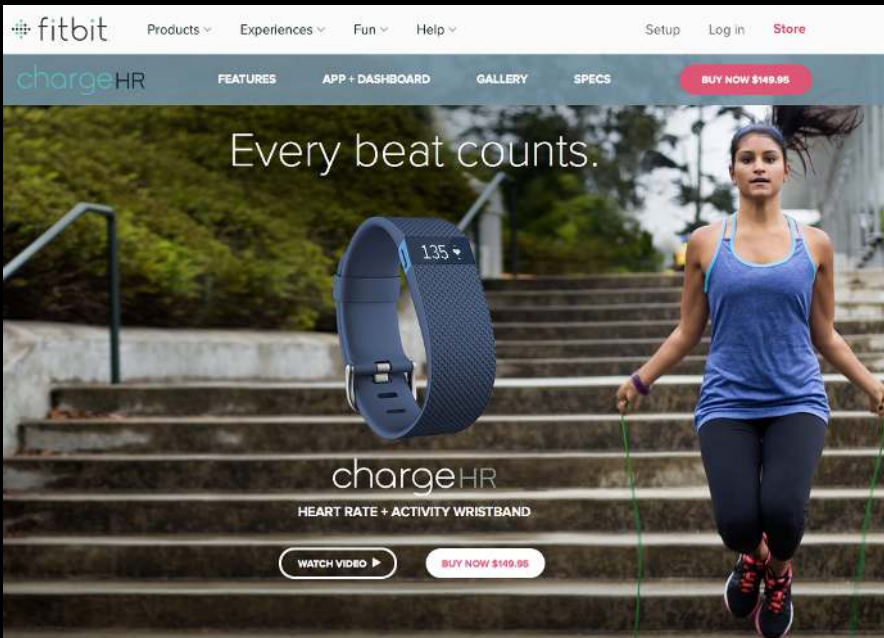
Create Account

If you do not have an Activation Code:

Request to Sign Up

UCLA Health Victim of a Criminal Cyber Attack

UCLA Health is sending letters to



The Fitbit Charge HR website features a clean, modern design. The header includes the Fitbit logo and navigation links for Products, Experiences, Fun, Help, Setup, Log in, and Store. Below the header, a blue bar contains the text "chargeHR" and navigation links for FEATURES, APP + DASHBOARD, GALLERY, and SPECS. A prominent pink button labeled "BUY NOW \$149.95" is positioned to the right. The main banner displays a large image of a woman running on a set of stairs, with a Fitbit Charge HR wristband shown in the foreground. The text "Every beat counts." is displayed above the wristband. Below the wristband, the text "chargeHR" and "HEART RATE + ACTIVITY WRISTBAND" are displayed. At the bottom, there are two buttons: "WATCH VIDEO" and "BUY NOW \$149.95".

fitbit

Products Experiences Fun Help

Setup Log in Store

chargeHR

FEATURES APP + DASHBOARD GALLERY SPECS

BUY NOW \$149.95

Every beat counts.

chargeHR

HEART RATE + ACTIVITY WRISTBAND

WATCH VIDEO

BUY NOW \$149.95

Internet of Things

Coherent, engaging experiences coordinated over time and space.



BRINGING INNOVATION TO THE NFL

ZEBRA MOTIONWORKS REAL-TIME PLAYER TRACKING SYSTEM

2014 NEXT GEN STATS BY THE NUMBERS

2,000+
NFL PLAYERS TAGGED AND TRACKED



5,800+
ZEBRA MOTIONWORKS TAGS DEPLOYED

TO PLAYERS, OFFICIALS & EQUIPMENT (SHOULDER PADS, YARD MARKERS, PYLONS)

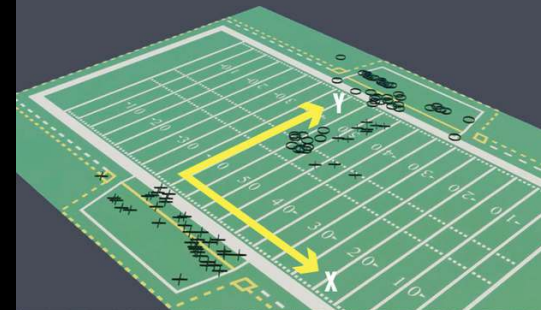


17,000+
PLAYS OF NEXTGEN STATS MEASURED



1,692,000,000+
SETS OF XY PLAYER COORDINATES

FOR PLAYER LOCATION MEASURED, TRANSMITTED AND STORED DURING NFL GAMES



68+ BILLION
BYTES OF PLAYER POSITION DATA

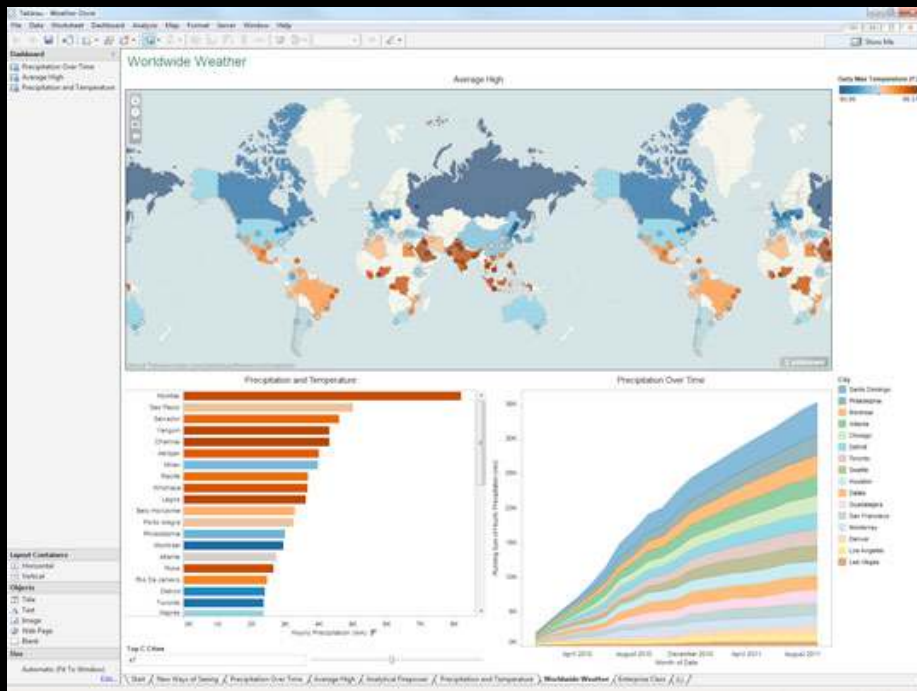
NOW THAT'S A LOT OF PLAYER DATA!



OFFICIAL ON-FIELD PLAYER TRACKING PROVIDER
OF THE NATIONAL FOOTBALL LEAGUE

Big Science

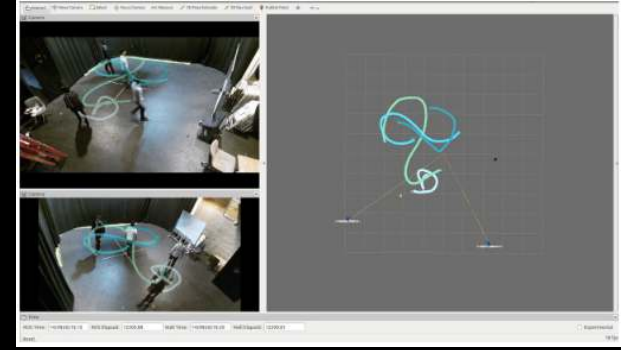
Exchange, provenance management, and public communication of huge amounts of data.



Grand Challenges

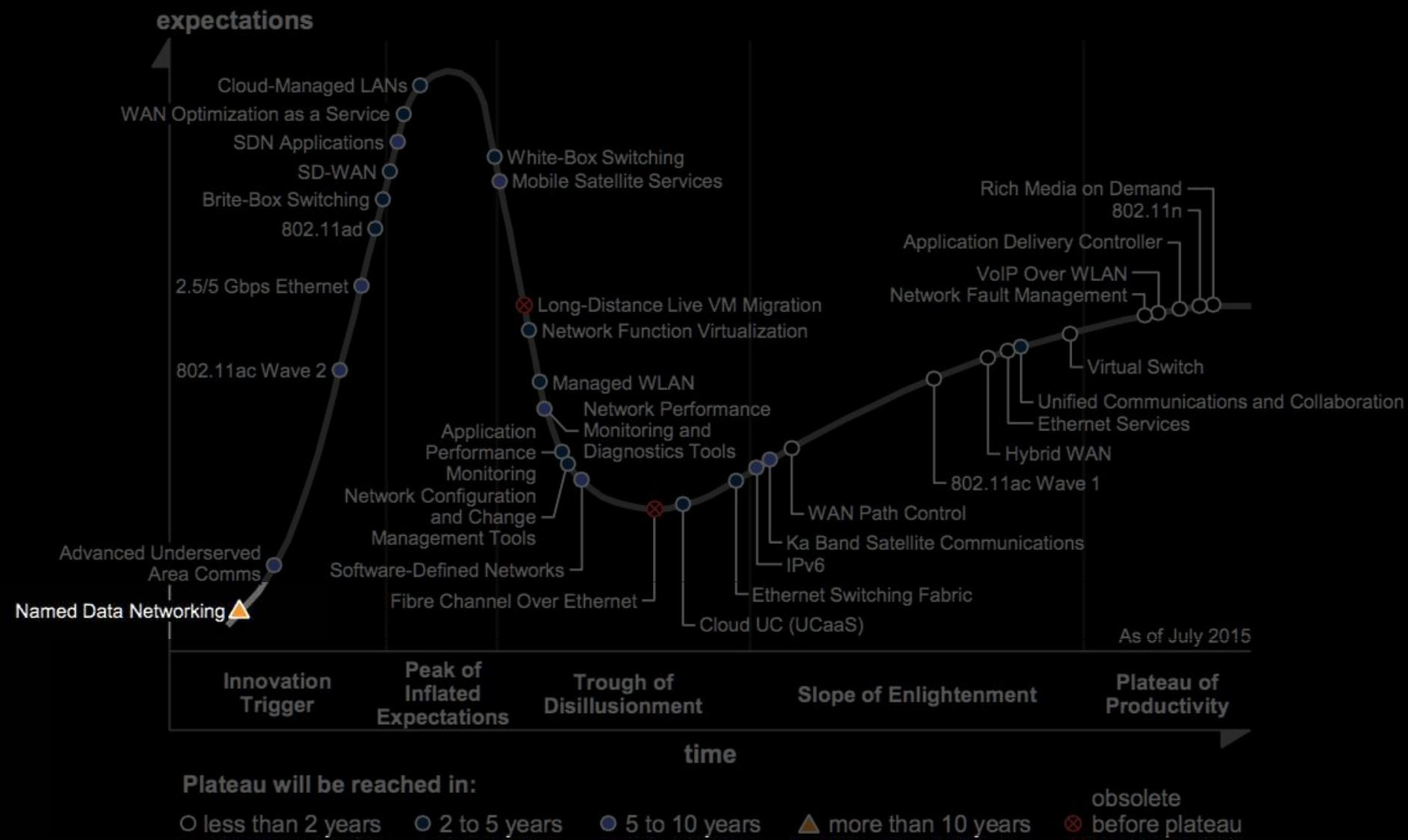
- Clouds, Circulation & Climate Sensitivity
- Melting Ice & Global Consequences
- Climate Extremes
- Regional Sea-level Change & Coastal Impacts
- Water Availability

Storytelling



But aren't those the edges? The fringe?

Figure 1. Hype Cycle for Networking and Communications, 2015



Source: Gartner (July 2015)

If we are asking people to look ahead in networking,
let's look ahead to the applications that are coming.

At the edges

For example,

- Multimedia experiences are created by designers and storytellers based on available technologies.
- Occasionally their goals directly motivate technologies.
- What motivates designers, storytellers, inventors, entrepreneurs...?
- The edges: of aesthetics, storytelling, interaction, design, etc.



What do the edges do?

Reformulate the Problem

“It is widely accepted that creative design is not a matter of first fixing the problem and then searching for a satisfactory solution concept; instead it seems more to be a matter of **developing and refining together both the formulation of the problem and ideas for its solution...**”

- Cross & Dorst (1999), quoted by Brooks (2010)

Example:

Streaming video content will not stay static

engadget

YouTube's latest experiment: multiple camera angles

by Richard Lawler | @rjcc | February 4th 2015 At 2:33pm



Sports broadcasts were among the first to give viewers the chance to pick how they watch the action on the internet, but now YouTube is giving the feature a try. A video of artist Madiyn Bailey performing at YouTube Music Night lets users click through a choice of four camera angles as they watch, without



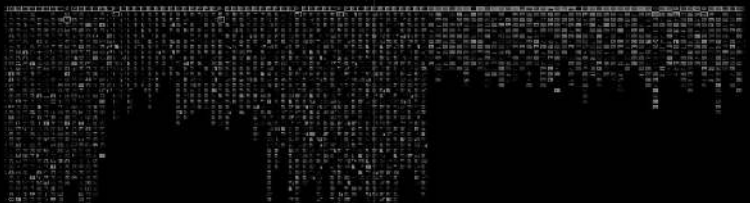
Maze Runner (2014), Fox

THE JOHNNY CASH PROJECT

Explore ▾ Contribute About Credits REGISTER / SIGN IN



TIME: 00:00:19 FRAME: 101



The Wilderness Downtown

Los Atlantis (UCLA, 2015)

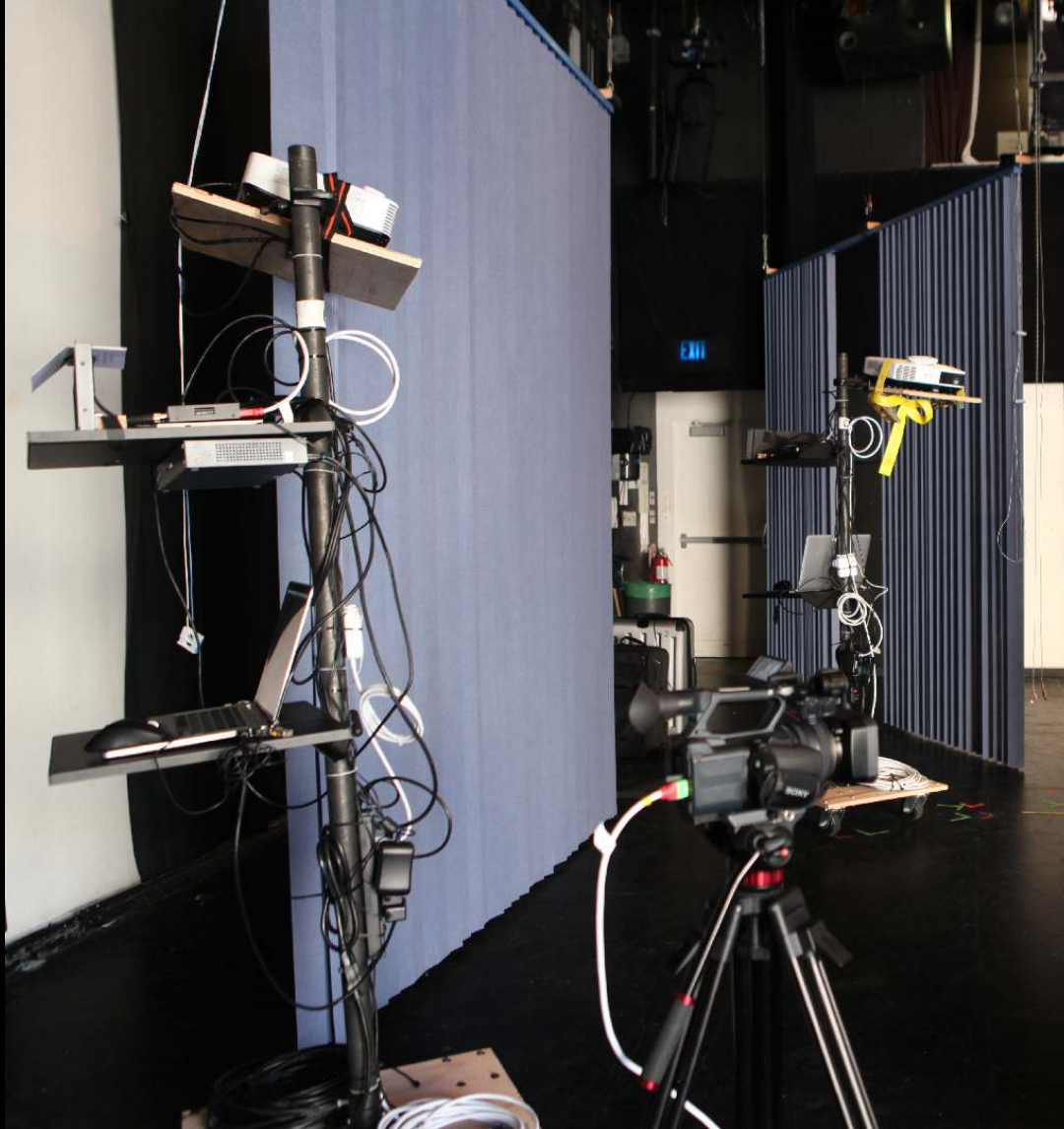


NDN Thought Experiment:

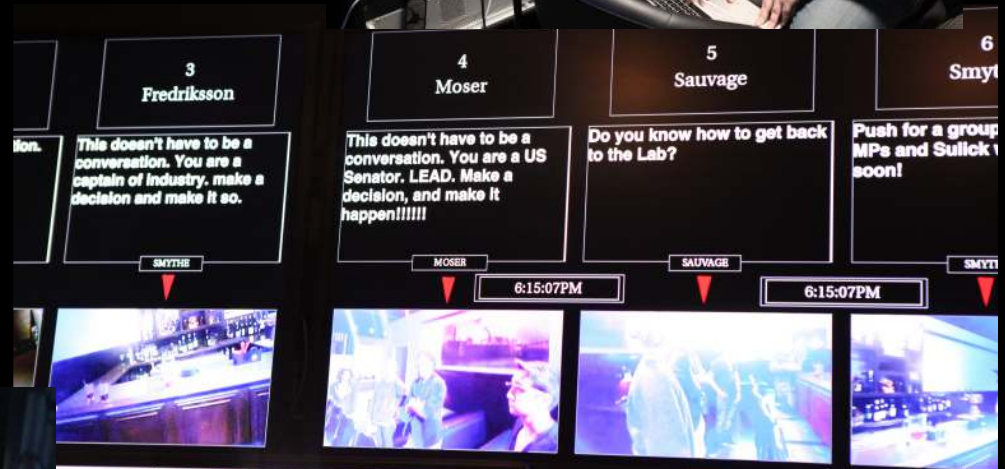
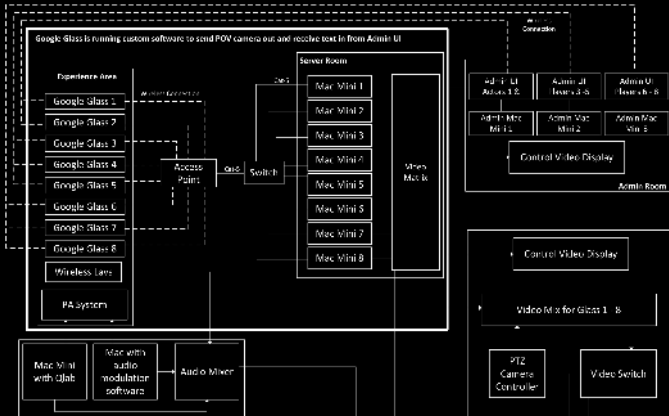
How to best enable just-in-time client-side
composability of streaming media elements?

“Expeditionary Networks”
are more common than you think

4 x Mobile Projector Tower
3-16Mbit/s down, 6+Mbit/s up ea.
NDN control



Grace Plains: Live-action Role Playing with Google Glass (2014)



NDN Thought Experiment:

Is IP-related brittleness holding us back
from making more *expeditionary* networks?

Going forward: What does this mean for NDN?

On NDN, app reformulation starts with the design of its namespace.

Storytelling apps from this year already using hierarchical data:

- Scene graphs for streaming 3D (Game engines)
- Processing networks for streaming 2D (Touchdesigner)
- App-level protocols in experimental multimedia (Open Sound Control)
- Message bus protocols (Robot Operating System)

But, mapping namespaces directly may not provide all the benefits it could...

- => Tussle between app, security, network requirements for naming is really challenging.
- => Efficient mechanisms for exchanging named data motivate more work on namespace synchronization as a higher level transport.

“It is widely accepted that creative design is not a matter of first fixing the problem and then searching for a satisfactory solution concept; instead it seems more to be a matter of **developing and refining together both the formulation of the problem and ideas for its solution...**”

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